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CHEMISTRY

Series 1-11

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SERIES 1

SECTION A (30 MARKS)

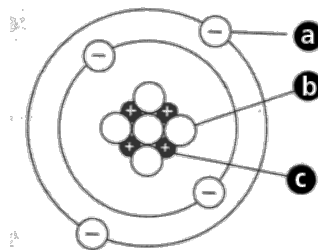
Answer *ALL* questions in this section.

1. In Oltanki Senior School laboratory, learners observed a certain atomic models. Three of the learners described it as follows

Peter: “The center contains protons.”

Achieng: “Electrons move around it.”

Kamau: “It is Rutherford’s model.”



- (a) Name the part labelled [+]. (1 mark)

_____.

- (b) Identify the particle labelled e^- . (1 mark)

_____.

- (c) State ONE feature of Rutherford’s atomic model. (1 mark)

- (d) Define the term isotope. (1 mark)

- (e) State ONE use of isotopes. (1 mark)

2. A Grade 10 learners from Chewoyet senior school tested the PH of five substances A, B, C, D and E using the universal indicator. He recorded the results in the table below.

Substance	PH
A	10
B	14
C	1
D	7
E	4

- a) Which of the substances is likely to be: (4mks)

i) Sodium hydroxide: _____.

ii) Pure water: _____.

ii) Tomato juice: _____.

iv) hydrochloric acid: _____.

(b) State ONE product of acid + metal reaction. (1 mark)

(c) Distinguish strong and weak acids. (2 marks)

3. The grid given below represents part of the periodic table. Study it and answer the questions that follow. The letters are not the actual symbols of the elements.

						A
B			G		H	E
	J		I	L		C
D						M
Y						

i. What name is given to the family of elements to which **A** and **C** belong? : _____ (1 mrk)

ii. Write the chemical formula of the sulphate of element **D**. : _____ (1 mrk)

iii. Which letter represents the most reactive (2mrks)

a. Metal: : _____

b. Non-metal : _____

iv. Select one element that belongs to period 4. : _____ (1mrk)

v. Explain why the Ionic radius of element **E** is bigger than the atomic radius. (2mrks)

vi. The electron configuration of a divalent anion of element **N** is 2.8.8. Indicate the position of element **N** on the periodic table drawn above. (1mrk)

vii. How do the atomic radii of **I** and **C** compare. Explain. (2mrks)

viii. Explain the trend in the 1st ionization energies of the elements **J**, **I** and **L**. (1mrk)

4. At **Malava Girls Senior School**, learners modeled bonding types.

a. Draw a diagram to show bonding in an ammonium ion. (N = 7, H = 1) (2 marks)

b. Identify the bond in NaCl. _____ (1 mark)

c. Name ONE property of ionic compounds. (1 mark)

d. Distinguish ionic and covalent bonds. (2 marks)

e. State ONE use of metallic bonding materials. (1 mark)

SECTION B: (50 MARKS)

5. A student in **Starehe Boys' Centre** prepared a salt using acid and base. They conducted the following reaction



(a) Name the salt formed. (1 mark)

(b) State the type of reaction. (1 mark)

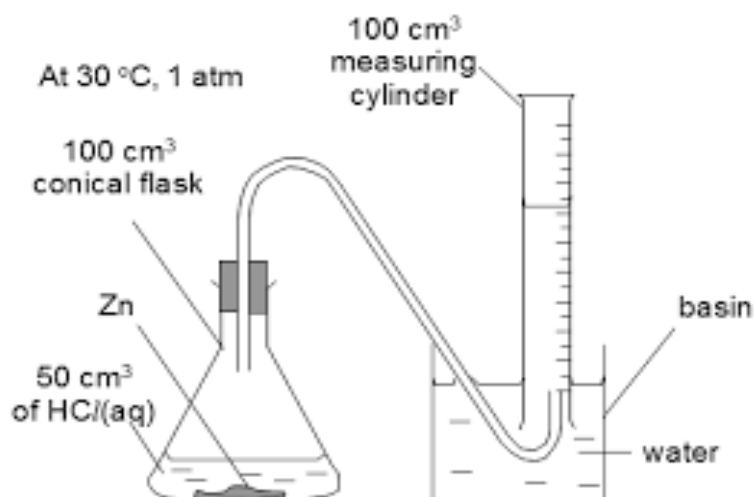
(c) Identify TWO types of salts. (2 marks)

i. _____

ii. _____

(d) State ONE use of salts. (1 mark)

6. At **Kakamega Senior School**, learners tested hydrochloric acid with metals.



$\text{Zn} + \text{HCl} \rightarrow \text{Gas bubbles observed}$

(a) Identify the gas produced. _____ (1 mark)

(b) Describe TWO properties of acids. (2 marks)

i. _____

ii. _____

(c) Write TWO reactions of acids with:

i. metal oxides (2 marks)

ii. carbonates (2 marks)

(d) State ONE use of acids in industry. (1 mark)

7. (a) The table shows the results of adding three metals, P, Q, and R, to dilute hydrochloric acid and to water.

metal	dilute hydrochloric acid	water
P	hydrogen produced	hydrogen produced
Q	no reaction	no reaction
R	hydrogen produced	no reaction

What is the order of reactivity of the metals? (1 mark)

(b) Study the information provided:-

Element	Atomic radius (nm)	Ionic radius (nm)	Melting point of oxide (°C)
W	0.381	0.418	-117
Y	0.733	0.669	849
Z	0.544	0.489	1399

Explain why the melting point of the oxide of W is lower than that of the oxide of Z (2 marks)

8. A learner in **Mary Hill Senior School** prepared copper sulfate crystals.

(a) Describe THREE methods of preparing salts. (3 marks)

- i. _____
- ii. _____
- iii. _____

(b) Explain behavior of salts in air. (2 marks)

- i. _____
- ii. _____

(c) State TWO uses of salts. (2 marks)

- i. _____
- ii. _____

9. The table below gives information about ions of **P** and **Y**

Ion	P ⁺	Y ²⁺
Electron arrangement	2.8	2.8.8
Number of Neutrons	12	16

a. Write the electron configuration for the atom of **Y** (1mk)

b. How many protons are there in the nucleus of (2mks)

(i) **P** _____

(ii) **Y** _____

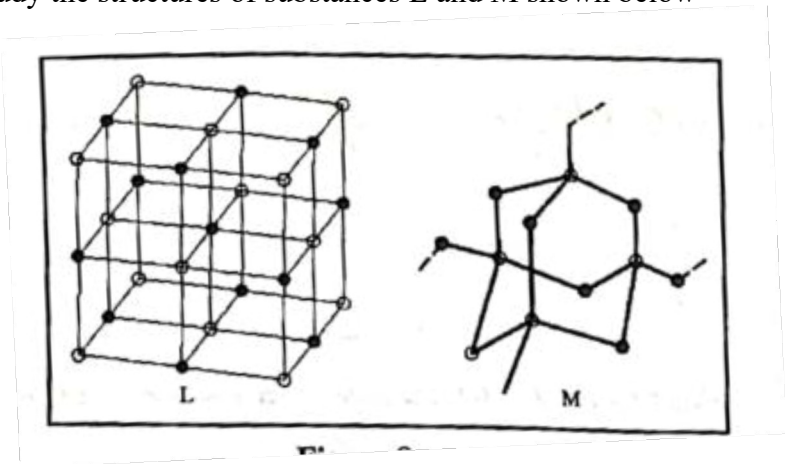
c. Write the formula of the compound formed when **P** and **Y** reacts (2mks)

(d) State TWO uses of sodium. (2 marks)

- i. _____
- ii. _____

(e) Identify ONE safety precaution when handling alkali metals. (2 marks)

10. Study the structures of substances L and M shown below



Complete **the table given** by giving the names that describe the type of structure and bonding in each substance. (2 marks)

Substances	L	M
------------	---	---

Structure		
Bonding		

11. The atomic numbers of nitrogen and fluorine are 7 and 9 respectively.

a. Draw electron dot (•) and cross (x) diagrams to illustrate bonding in the molecules of:

i. nitrogen; (1 mark)

ii. fluorine (1 mark)

b. With reference to the diagrams in (a), state why fluorine is more reactive than nitrogen. (1 mark)

12. At **Maranda Senior School**, learners studied chlorine isotopes.

Cl-35 = 75%

Cl-37 = 25%

(a) Define relative atomic mass. (2 marks)

(b) Calculate relative atomic mass of chlorine. (2 marks)

13. A factory in **Nairobi Industrial Area** produces fertilizers and detergents.

(a) Explain THREE uses of chemistry in daily life. (3 marks)

i.

ii.

iii.

(b) State TWO environmental effects of chemical misuse. (2 marks)

i.

ii.

(c) Suggest TWO safety measures in chemical handling. (2 marks)

i.

ii.

14. Ethanol is a liquid at room temperature but does not conduct electricity. Explain.(2 marks)

SERIES 2

Answer all questions in this section.

1. A mother is seen reading labels on detergent containers before buying.

(a) Define Chemistry. (1mk)

(b) Identify one branch of chemistry. (1mk)

(c) State one career related to the branch you named in (b). (1mk)

2. A clinic issues medicine to patients in clearly labelled packets.

(a) Define the term drug. (1mk)

(b) What is a prescription? (1mk)

(c) State the meaning of dosage. (1mk)

(d) Identify one consumer right when receiving prescribed drugs. (1mk)

3. Name two effects of abusing the following drugs:

(a) Cigarettes (1mk)

(b) Khat (miraa) (1mk)

4. State three roles of chemistry in modern agriculture. (3mks)

- i. _____
- ii. _____
- iii. _____

5. A student was asked to name the first 20 elements in order.

Write the names of elements with atomic numbers:

(a) 1 (1mk)

(b) 8 (1mk)

(c) 12 (1mk)

6. State two differences between atomic number and mass number. (2mks)

7. Give three responsibilities of learners in maintaining a safe learning environment. (3mks)

- i. _____
- ii. _____
- iii. _____

8. A medicine advert states:

“This drug may cause side effects such as dizziness.”

(a) What is a substance? (1mk)

(b) Give two common routes of drug intake. (2mks)

(c) State one side effect of heroin abuse. (1mk)

9. State three contributions of chemistry in the entertainment industry. (3mks)

10. Write the electronic configuration using s and p notation for:

(a) Nitrogen (Atomic number = 7) (1mk)

(b) Magnesium (Atomic number = 12) (1mk)

(c) Argon (Atomic number = 18) (1mk)

SECTION B (60 MARKS)

Answer ALL questions.

11. A community hospital receives complaints that some herbal drinks sold around the market are causing dizziness and stomach upsets.

(a) Explain three roles of chemistry in ensuring safety of such herbal products. (6mks)

- i.
- ii.
- iii.

(b) Describe two responsibilities of consumers when buying herbal substances. (2mks)

(c) State two qualities of a safe learning/working environment in a chemistry lab. (2mks)

12. A youth centre conducts research on drugs abused by teenagers in the area.

(a) Explain two reasons why young people abuse drugs such as glue/cobbler’s blue. (4mks)

(b) For each of the drugs below, describe one mode of intake and one health effect:

(i) Cocaine (2mks)

(ii) Illicit brew (chang’aa) (2mks)

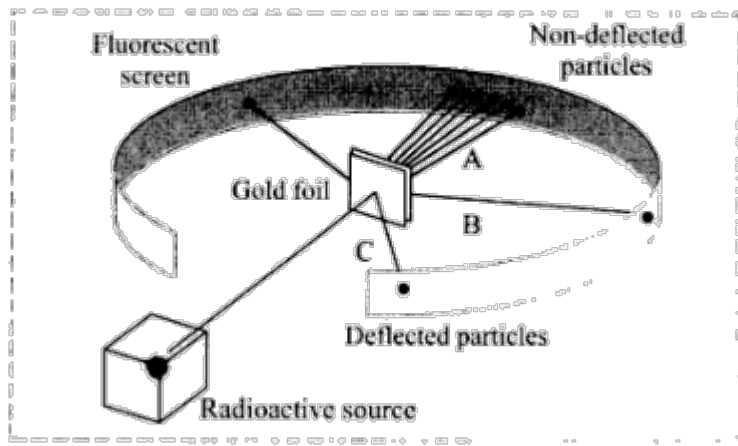
(iii) Cannabis (bhang) (2mks)

(c) State two long-term effects of mandrax on the human body. (2mks)

13. A teacher uses models to explain how scientists discovered the structure of the atom.

(a) State two postulates of Dalton’s atomic theory. (2mks)

(b) With reference to the representative model of the gold foil experiment, answer the following question



What evidence regarding an atom do lines A, B and C provide? (3mks)

(c) Give two conclusions made from the experiment. (2mks)

(d) Define the following terms:

(i) Isotope (1mk)

(ii) Orbital (1mk)

(e) Identify the subatomic particle present in different numbers in isotopes of the same element. (1mk)

14. A chemist is analysing an element, X, whose isotopes have masses 20 and 22, in the ratio 3:1.

(a) Calculate the relative atomic mass of element X. (3mks)

(b) Draw a labelled diagram showing the first three energy levels in an atom. (3mks)

(c) Explain the relationship between energy levels and orbitals. (3mks)

(d) State the order of filling of electrons into the first four orbitals. (2mks)

(e) Write the electron configuration using s and p notation for:

(i) Phosphorus ($Z = 15$) (2mks)

(ii) Neon ($Z = 10$) (1mk)

15. A career guidance teacher organizes a talk on chemistry-related professions.

(a) List four branches of chemistry. (4mks)

(b) Write a career to the correct branch of chemistry: (4mks)

(i) Pharmacist: _____

(ii) Petroleum engineer: _____

(iii) Food technologist: _____

(iv) Forensic scientist: _____

(c) Explain three ways chemistry is applied in the manufacturing industry. (6mks)

SERIES 3

SECTION A: Theory (70 Marks)

Answer all questions in this section.

1. A Grade 10 learner of Mang'u Senior School notices that fertilisers improve crop yields, medicine cure diseases, fuel produces energy and plastics are used in daily life.

a) Using the above scenario: Define **Chemistry** as a field of science. (2 marks)

b) Explain **four roles of Chemistry in daily life** shown in the scenario. (4 marks)

i) _____

ii) _____

iii) _____

iv) _____

b) State **three reasons** why learning Chemistry is important to society. (3 marks)

i) _____

ii) _____

iii) _____

2. During a career guidance session, learners research different careers related to Chemistry using books and the internet.

a) Name and describe **four branches of Chemistry**. (8 marks)

i) _____

ii) _____

iii) _____

iv) _____

b) State **three career opportunities** related to Chemistry. (3 marks)

i) _____

ii) _____

iii) _____

3. Grade 10 learners of Kilgoris Boys Senior School discovered that Chemistry plays a major role in different industries such as agriculture, pharmaceuticals, food processing, energy and entertainment.

a) Explain **four applications of Chemistry** in the following areas: (8 marks)

	Area	Application of chemistry
i.	Agriculture	
ii.	Pharmaceutical industry	
iii.	Manufacturing industry	
iv.	Entertainment industry	

b) State **two benefits** of Chemistry to economic development. (2 marks)

- i) _____
- ii) _____

4. A health officer warns learners about the dangers of drug and substance abuse in schools.

a) Define the following terms:

i) Drug (1 mark)

ii) Prescription (1 mark)

iii) Dosage (1 mark)

b) Explain **three effects of drug and substance abuse** on learners. (6 marks)

- i) _____
- ii) _____
- iii) _____

c) State **one responsibility** of learners in promoting a safe and healthy learning environment. (1 mark)

5. Learners at Litein Girls Senior School watched a simulation of Rutherford's Gold Foil experiment during a Chemistry lesson.

a) Describe **two observations** made in Rutherford's experiment. (2 marks)

i)

ii)

b) State **two conclusions** drawn from the experiment. (2 marks)

i)

ii)

c) Name **two sub-atomic particles** found in an atom. (2 marks)

i)

ii)

6. The table below shows PH values of solutions A, B, C and D.

Solution	PH
A	3.0
B	13.0
C	8.5
D	7.0

(a) Identify a solution which is. (2 marks)

(i) Strongly acidic. ($\frac{1}{2}$ mark)

(ii) Strongly basic ($\frac{1}{2}$ mark)

7. A teacher at Maranda Senior School stated that Element X has two isotopes:

- i. X_1 has mass number 35 and abundance 75%
- ii. X_2 has mass number 37 and abundance 25%

a) Define:

i) Atomic number (1 mark)

ii) Mass number (1 mark)

b) Calculate the **relative atomic mass** of element X. (4 marks)

c) Define the term **isotopes**. (2 marks)

d) State **two uses of isotopes**. (2 marks)

i) _____

ii) _____

8. The grid below shows part of the periodic table. Use it to answer question that follow. The letters do not represent actual symbols.

			S	U	V
P	R		T		W
Q					

(a) Give the name of: (2 marks)

(i)Element S

(ii)Element W

9. The electron arrangement of an element Y is:

2.8.8.1

a) Identify the **period** and **group** of element Y. (2 marks)

i. Period: _____.

ii. Group: _____.

b) Identify the element. (1 mark)

_____.

c) State the **chemical family** to which element Y belongs. (1 mark)

d) Predict the **type of ion** formed by element Y and explain why. (2 marks)

d) State **two reasons** why noble gases are chemically stable. (2 marks)

i) _____

ii) _____

e) Give an example of an element under the following category. (3 marks)

	Category	Element
i.	<i>Alkaline earth metal</i>	
ii.	<i>Halogen</i>	
iii.	<i>Noble gas</i>	

10. a) Write the **electron arrangement (s and p notation)** for the following ions:

i) Na^+ (2 marks)

ii) O^{2-} (2 marks)

b) Using valency or oxidation numbers, write the chemical formula of:

i) Magnesium chloride (2 marks)

ii) Aluminium oxide (2 marks)

c) Balance the following chemical equation:



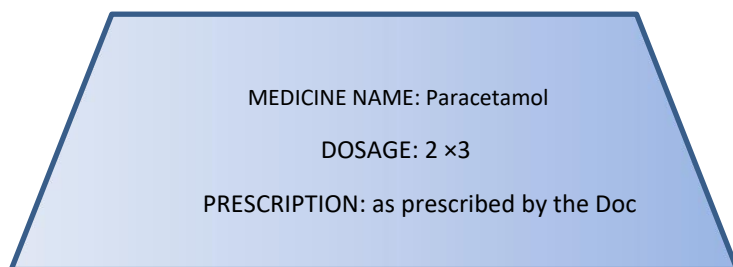
SERIES 4

1. Define the term Chemistry. (1 mark)
2. State two reasons why Chemistry is called a field of science. (2 marks)
3. List three careers related to the field of Chemistry. (3 marks)
4. Name the branch of Chemistry to the correct description. (4 marks)

Description	Branch
A. Study of energy changes	
B. Study of carbon compounds	
C. Study of metals and salts	
D. Identifying substances	

5. (a) State three ways Chemistry is applied in agriculture. (3 marks)
- (b) Mention the relationship between chemistry and these science fields (3 marks)
 - i. Biology
 - ii. Physics
 - iii. Community Service Learning

6. The diagram below represents a medicine label.



- (a) What is meant by the term **dosage**? (1 mark)
 - (b) What is meant by the term **prescription**? (1 mark)
 - (c) State **one danger** of ignoring dosage instructions. (1 mark)
7. State two differences between a drug and a substance. (2 marks)

<i>Drug</i>	<i>Substance</i>
-------------	------------------

8. Give three examples of substances that are commonly abused in society. (3 marks)

9. (a) Explain three effects of drug and substance abuse on a learner. (3 marks)

(b) Suggest **two ways** the school can help prevent such misuse. (2 marks)

10.

(a) Explain what is meant by **gender stereotyping**. (2 marks)

(b) State **two ways** gender stereotyping affects learners' career choices. (2 marks)

(c) Suggest **two ways** schools can reduce gender stereotyping in science careers. (2 marks)

11. (a) Define the term atom. (1 mark)

(b) State three particles found in the nucleus of an atom. (3 marks)

12. Draw a diagram showing the Rutherford's atomic model (2 marks)

13. Distinguish between:

(a) Atomic number and mass number. (2 marks)

(b) Protons and electrons. (2 marks)

14. An element has atomic number 12 and mass number 24.

i. State the number of protons. (1 mark)

ii. State the number of neutrons. (1 mark)

iii. State the number of electrons. (1 mark)

15. State two similarities between Dalton's and Rutherford's atomic models. (2 marks)
16. Argon has three isotopes argon- 36, argon- 38 and argon-40. The abundance of each is 0.337, 0.163 and 99.6 respectively. Calculate the relative atomic mass of argon. (3marks)
17. Oxygen has two isotopes; oxygen-16 and oxygen-18. If the RAM of oxygen is 16.2, find the percentage abundance of each (3marks)
18. Write the electron arrangement of the following elements using the s and p notation (4marks)
- Carbon
 - Sodium
 - Sulphur
 - Calcium
19. State two rules followed in orbital filling. (1 mark)
20. Define the term periodic table. (1marks)
21. State two reasons why the periodic table is important in Chemistry. (2 marks)
22. State four chemical families found in the periodic table. (4 marks)
23. Draw the periodic table for the first 20 elements indicating the correct symbols in the boxes. Indicate the periods and groups (4 marks)
24. Mention four transition elements (4 marks)
25. Define a radical. (1 mark)
26. Draw the electron spins for the atoms of the following elements(2 marks)

Magnesium

Potassium

27. Write the electron arrangement of the following ions using s and p notation (2 marks)

Fluorine ion

Aluminium ion

28. Use valencies to write the formulae of the following compounds: (5 marks)

a. Magnesium oxide:

b. Calcium chloride:

c. Aluminium oxide:

d.

e. Potassium nitrate:

f. Calcium carbonate:

29. Using dots and crosses draw the ions of

i. Magnesium (2 marks)

ii. Chlorine (2 marks)

30. Write balanced chemical equations including the physical states for the following (10 marks)

a) Copper + Oxygen $\longrightarrow$ Copper (II) Oxide

b) Zinc + Hydrochloric Acid $\longrightarrow$ Zinc Chloride + Hydrogen Gas

c) Calcium + Water $\longrightarrow$ Calcium Hydroxide + Hydrogen Gas

d) Carbon + Oxygen $\longrightarrow$ Carbon (II) Oxide

e) Sodium + Water $\longrightarrow$ Sodium Hydroxide + Hydrogen Gas

SERIES 5

SECTION A (30 MARKS)

Answer ALL questions in this section.

1. The diagram below represents an atom of carbon.



Identify: (2 marks)

- a) Atomic number – _____
b) Mass number – _____

2. Naturally occurring magnesium exists as different isotopes. Use the data provided to calculate the relative atomic mass. (3 marks)

Isotope	Relative Abundance (%)	Mass of Isotope
${}^{24}\text{Mg}$	78.99	23.98
${}^{26}\text{Mg}$	10.00	24.98
${}^{27}\text{Mg}$	11.01	25.98

3. Sodium has an atomic number of 11. Write its electron configuration in s and p notation. (3 marks)

4. Brian studied the contributions of early chemists. State one contribution of **Antoine Lavoisier** to chemistry. (2 marks)

5. Given that the sulfite ion is SO_3^{2-} , state its valency. (1 mark)

6. Learners researched the chemical formula of ammonium sulphate. Write the correct formula. (1 mark)

7. Write a balanced chemical equation for the reaction between magnesium and steam. (2 marks)

8. Kevin learned about chemical bonding. Explain how the following bonds form: (3 marks)

a) Ionic bond

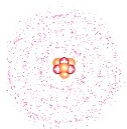
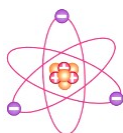
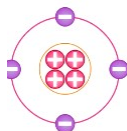
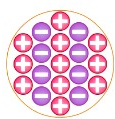
b) Covalent bond

c) Metallic bond

9. According to atomic theory, energy levels have specific capacities. Fill in the blanks: (2 marks)

- i. First energy level – _____ electrons
- ii. Second energy level – _____ electrons
- iii. Third energy level – _____ electrons

10. Scientists developed various atomic models. Match them to their pioneer chemist



1904
J.J.
Thomson

1803
John
Dalton

1911
Ernest
Rutherford

1926
Erwin
Schrodiner

1913
Niels
Bohr

11. A _____ is the smallest particle of an element that participates in a chemical reaction. (1 mark)

12. Government agencies regulate product quality. State one role played by such agencies in Kenya. (1 mark)

13. Drug misuse can have harmful effects. State two health effects of drug abuse. (2 marks)

- i. _____
- ii. _____

14. **Samuel** is interested in industrial chemistry. State two careers related to this field. (2 marks)

- i. _____
- ii. _____

15. Chemistry is widely applied in medicine production. Explain one application in the pharmaceutical industry. (1 mark)

16. Biochemistry is an important branch of chemistry. State what it deals with. (1 mark)

17. **Aisha** was asked to define Chemistry. State the meaning of Chemistry. (2 marks)

SECTION B (50 MARKS)

Answer ALL questions in the spaces provided.

18. During a health talk, learners discussed drug and substance use.

a) Define: (4 marks)

i) Drug

ii) Substance use

b) State two academic effects of drug abuse. (2 marks)

- i. _____
- ii. _____

19. A consumer examined a product label before purchase.

a) State three components of a label. (3 marks)

- i. _____
- ii. _____
- iii. _____

b) Explain the importance of proper labelling. (2 marks)

- i. _____
- ii. _____

c) State two roles of quality assurance bodies in Kenya. (2 marks)

- i. _____
- ii. _____

20. Learners studied atomic structure concepts.

a) Define atomic number. (2 marks)

b) Define isotopes. (2 marks)

c) An atom has mass number 27 and atomic number 13:

i) Number of neutrons (2 marks)

ii) Number of electrons (2 marks)

21. Students practised electron configurations.

a) Write electron arrangement for: (2 marks)

i) Sodium

ii) Chlorine

b) Write s and p notation for calcium. (3 marks)

22. The teacher gave out the electron configurations of sulphur and potassium to learners and asked them to determine their positions in the periodic table.

a) Given that sulphur has the configuration, $1s^2 2s^2 2p^6 3s^2 3p^4$:

i) Identify its period. (1 mark)

ii) Identify its group. (1 mark)

b) Potassium has the configuration, $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$:

i) Identify its period. (1 mark)

ii) Identify its group. (1 mark)

c) Name the chemical family of potassium. (1 mark)

23. Learners studied ion formation.

a) State the octet rule. (2 marks)

b) Distinguish between cation and anion. (2 marks)

24. Electron arrangement influences chemical behaviour.

a) Explain its role in the periodic table. (1 mark)

b) Explain why elements in the same group have similar properties. (1 mark)

c) Explain how it helps in writing chemical formulae. (2mark)

25. Learners discussed the importance of chemistry in daily life.

a) Outline four applications of chemistry. (4 marks)

- i. _____
- ii. _____
- iii. _____
- iv. _____

b) Name two industries where chemistry is applied. (2 marks)

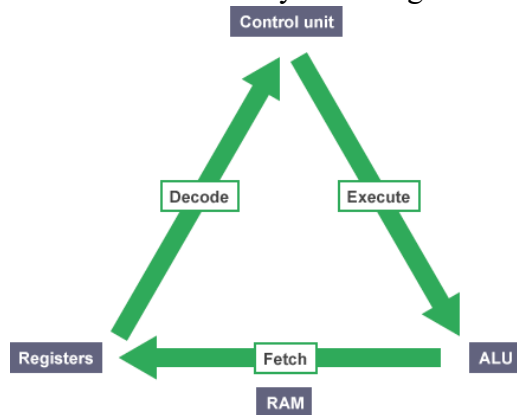
- i. _____
- ii. _____

SERIES 6

SECTION A (30 MARKS)

Answer ALL questions in this section.

1. **Brian** illustrated the fetch-execute cycle during a lesson. Explain how the fetch-execute cycle works. (2marks)



i. _____

2. **Lydia** identified a printer as an output device. Explain why it is classified as an output device. (1 mark)

3. Classify the following input devices: (2 marks)



a) _____
d) _____

b) _____

c) _____

4. Give two examples of visual output devices. (2 marks)

i. _____

ii. _____.

5. During research, learners identified a product with the feature shown below.



Name this feature. (1 mark)

_____.

6. Distinguish between RAM and ROM. (2 marks)

RAM	ROM

7. Mr Mwangi was given a laptop by The Good People International. He wants to install an O.S. Identify the following operating systems he can choose from: (4 marks)



a) _____ b) _____ c) _____ d) _____

8. Match the following system buses with their functions: (3 marks)

- | | |
|----------------|--|
| a) Data bus | Carries the memory addresses that indicate where data or instructions are stored |
| b) Address bus | Transmits control signals from the Control Unit in the CPU to other components. |
| c) Control bus | Transfers data signals and instructions between components. |

9. Kevin a grade 10 learner, learned about processor design. What does RISC stand for? (1 mark)

10. State one difference between RISC and CISC based on instruction set. (2 marks)

RISC	CISC

11. Mama Timo bought the following computer for her grade 10 son



The computer shown belongs to which generation?

_____ (1 mark)

12. Learners at Victory Senior School explored different web browsing tools. Identify the following: (3 marks)



a) _____

b) _____

c)

13. State two features of fifth-generation computers. (2 marks)

- i. _____
- ii. _____.

14. Early computing devices laid the foundation of modern computers. Give two examples. (2 marks)

- i. _____
- ii. _____.

15. Daniel prefers using ATMs for transactions. Explain how computers are used in ATM machines. (2 marks)

- i. _____
- ii. _____.

SECTION B (50 MARKS)

Answer ALL questions in the spaces provided.

16. Learners from **Khallid Senior School** studied first-generation computers.

a) Identify the main technology used. (1 mark)

b) State two characteristics. (2 marks)

- i. _____
- ii. _____.

c) State two disadvantages. (2 marks)

- i. _____
- ii. _____.

17. Grade 10 learners of Alliance Senior School learned about computer architecture.

a) Name the scientist behind the Von Neumann architecture.

_____ (1 mark)

b) State two components of this architecture. (2 marks)

- i. _____
- ii. _____.

c) Explain functions of: *(4 marks)*

i) ALU

ii) Control Unit

d) State one function of registers. *(1 mark)*

18. During a practical lesson, learners were asked to investigate how computers receive data from input devices and produce results through output devices using different methods.

a) Describe the keying method used by input devices to enter data into computers. *(2 marks)*

i.

b) State two pointing devices. *(2 marks)*

i.

ii.

c) Explain two output methods. *(4 marks)*

i.

ii.

d) Give two audio output devices. *(2 marks)*

i.

ii.

19. Students studied cloud storage.

a) Define remote storage. *(1 mark)*

b) State two types of cloud storage. *(2 marks)*

i.

ii.

c) State two advantages. *(2 marks)*

i.

ii.

d) State two disadvantages. *(2 marks)*

i.

ii. _____.

20. Learners handled external storage devices.

a) Give two examples. (2 marks)

i. _____.

ii. _____.

b) Explain two factors to consider when selecting storage devices. (2 marks)

i. _____.

ii. _____.

c) State one safety precaution. (1 mark)

21. Learners studied CPU performance.

a) Define a core. (2 marks)

b) State two types of CPUs based on cores. (2 marks)

i. _____.

ii. _____.

c) State two advantages of multi-core CPUs. (2 marks)

i. _____.

ii. _____.

d) State two characteristics of single-core CPUs. (2 marks)

i. _____.

ii. _____.

22. Learners researched modern technologies.

a) Define Internet of Things (IoT). (2 marks)

b) State three advantages of cloud computing. (3 marks)

i. _____.

ii. _____.

c) Give two examples of IoT devices. (2 marks)

SERIES 7

SECTION A: Theory (80 Marks)

Answer all questions in this section.

1. a) Define **Chemistry** as a field of science. (2 marks)

- b) State **four branches of Chemistry**. (2 marks)

- i. _____.
- ii. _____.

- c) Identify **two careers related to Chemistry** relevant to Agriculture. (2 marks)

- i. _____.
- ii. _____.

- d) Give **two reasons** why Chemistry is important in farming. (2 marks)

- i. _____.
- ii. _____.

2. The diagram below shows areas where Chemistry is applied in daily life.



X



Y



Z

- a) Identify **three industries** shown above that use Chemistry. (3 marks)

X: _____.

Y: _____.

Z: _____.

b) Explain **three ways Chemistry contributes to development in Kenya**. (3 marks)

- i. _____.
- ii. _____.
- iii. _____.

c) State **one negative effect** of misuse of chemicals in daily life. (1 mark)

_____.

_____.

3. A farmer applies chemicals without following instructions on the label.

a) Distinguish between the following pairs of terms: (4 marks)

i) Drug and medicine

_____.

_____.

ii) Legal drugs and illegal drugs

_____.

_____.

b) Explain **three effects** of drug and substance abuse on an individual. (3 marks)

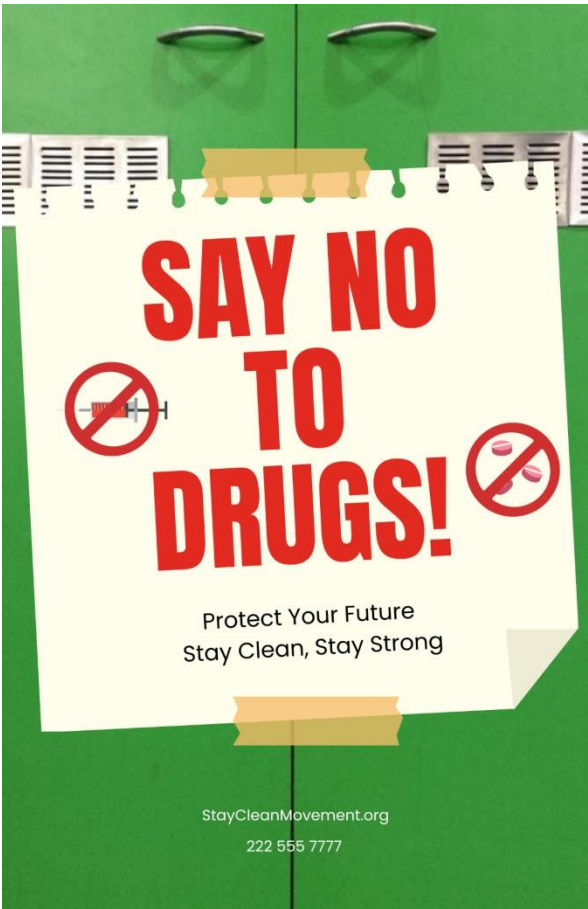
- i. _____.
- ii. _____.
- iii. _____.

c) State **one responsibility** of learners in promoting a safe and healthy learning environment. (1 mark)

_____.

_____.

4. Below is a poster displayed in a school laboratory.



a) Identify **two safety messages** shown in the poster. (2 marks)

- i. _____
- ii. _____

b) Explain **three consumer rights** related to drug prescription and chemical use. (3 marks)

- i. _____
- ii. _____
- iii. _____

c) State **two ways** students can sensitize the community about drug abuse. (2 marks)

- i. _____
- ii. _____

5. Grade 10 learners of Starehe Boys Centre drew diagrams to represents the **structure of an atom**.

a) Name **three sub-atomic particles** found in an atom. (3 marks)

- i. _____.
- ii. _____.
- iii. _____.

b) State **one charge** for each particle named in (a). (3 marks)

- i. _____.
- ii. _____.
- iii. _____.

c) Distinguish between: (4 marks)

i) Atomic number

ii) Mass number

6. An element X has the following isotopes:

Isotope	Mass Number	Abundance (%)
X ₁	35	75
X ₂	37	25

a) Define **isotopes**. (2 marks)

b) Calculate the **relative atomic mass** of element X. (2 marks)

c) Write the **electron arrangement** of element X using **s and p orbitals**.(2 marks)

7. The partial periodic table below shows elements A, B, C, and D.

A							B
X			C				
	D						

a) Identify the elements represented by the letters above: (4 marks)

A: _____

B: _____

C: _____

D: _____

b) Identify the **group and period** of element B. (2 marks)

Group	
Period	

c) Explain how the **position of an element** in the periodic table relates to its **electron arrangement**. (2 marks)

d) Predict the **type of ion** formed by element X and explain your answer. (2 marks)

8. A farmer mixes calcium hydroxide with ammonium sulphate fertilizer.

a) Define **valency** and **oxidation number**. (2 marks)

Valency: _____

Oxidation number: _____

b) Derive the **chemical formula** of ammonium sulphate. (2 marks)

c) Write a **balanced chemical equation** for a neutralization reaction involving calcium hydroxide. (3 marks)

9. Define the following terms: (2 marks)

i) Atom:

ii) Compound:

10. (a) Write the chemical symbols of the following elements: (3 marks)

i) Hydrogen: _____

ii) Oxygen: _____

iii) Iron: _____

(b) Write the names of the elements represented by the following symbols: (2 marks)

i) Na: _____

ii) K: _____

11. (a) State one use of each of the following metals: (2 marks)

i) Calcium:

ii) Iron:

12. Complete the table below: (8 marks)

Compound Name	Element 1	Element 2	Chemical Formula
Water	Oxygen		
Common salt	Sodium		
Carbon dioxide	Carbon		
Ammonia	Hydrogen		

13. Name the following compounds: (3 marks)

i) HCl: _____

ii) MgO: _____

iii) CO₂: _____

14. Complete the following reactions: (2 marks)

i) Sodium + Chlorine → _____

ii) Calcium + Fluorine → _____

SERIES 8

SECTION A (30 MARKS)

Answer ALL questions in this section.

1. The diagram below represents an atom of carbon.



Identify: (2 marks)

a) Atomic number – _____

b) Mass number – _____

2. Naturally occurring magnesium exists as different isotopes. Use the data provided to calculate the relative atomic mass. (3 marks)

Isotope	Relative Abundance (%)	Mass of Isotope
24Mg	78.99	23.98
26Mg	10.00	24.98
27Mg	11.01	25.98

3. Sodium has an atomic number of 11. Write its electron configuration in s and p notation. (3 marks)

4. **Brian** studied the contributions of early chemists. State one contribution of **Antoine Lavoisier** to chemistry. (2 marks)

5. Given that the sulfite ion is SO_3^{2-} , state its valency. (1 mark)

6. Learners researched the chemical formula of ammonium sulphate. Write the correct formula. (1 mark)

7. Write a balanced chemical equation for the reaction between magnesium and steam. (2 marks)

8. Kevin learned about chemical bonding. Explain how the following bonds form: (3 marks)

a) Ionic bond

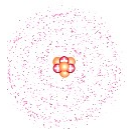
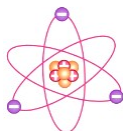
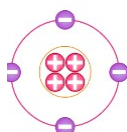
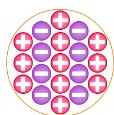
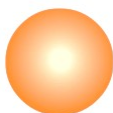
b) Covalent bond

c) Metallic bond

9. According to atomic theory, energy levels have specific capacities. Fill in the blanks: (2 marks)

- i. First energy level – _____ electrons
- ii. Second energy level – _____ electrons
- iii. Third energy level – _____ electrons

10. Scientists developed various atomic models. Match them to their pioneer chemist



1904
J.J.
Thomson

1803
John
Dalton

1911
Ernest
Rutherford

1926
Erwin
Schrodinger

1913
Niels
Bohr

11. A _____ is the smallest particle of an element that participates in a chemical reaction. (1 mark)

12. Government agencies regulate product quality. State one role played by such agencies in Kenya. (1 mark)

13. Drug misuse can have harmful effects. State two health effects of drug abuse. (2 marks)

- i. _____
- ii. _____

14. Samuel is interested in industrial chemistry. State two careers related to this field. (2 marks)

- i. _____
- ii. _____

15. Chemistry is widely applied in medicine production. Explain one application in the pharmaceutical industry. (1 mark)

16. Biochemistry is an important branch of chemistry. State what it deals with. (1 mark)

17. Aisha was asked to define Chemistry. State the meaning of Chemistry. (2 marks)

SECTION B (50 MARKS)

Answer ALL questions in the spaces provided.

18. During a health talk, learners discussed drug and substance use.

a) Define: (4 marks)

i) Drug

ii) Substance use

b) State two academic effects of drug abuse. (2 marks)

- i. _____

ii. _____

19. A consumer examined a product label before purchase.

a) State three components of a label. (3 marks)

i. _____

ii. _____

b) Explain the importance of proper labelling. (2 marks)

i. _____

c) State two roles of quality assurance bodies in Kenya. (2 marks)

i. _____

ii. _____

20. Learners studied atomic structure concepts.

a) Define atomic number. (2 marks)

b) Define isotopes. (2 marks)

c) An atom has mass number 27 and atomic number 13:

i) Number of neutrons (2 marks)

ii) Number of electrons (2 marks)

21. Students practised electron configurations.

a) Write electron arrangement for: (2 marks)

i) Sodium

ii) Chlorine

b) Write s and p notation for calcium. (3 marks)

22. The teacher gave out the electron configurations of sulphur and potassium to learners and asked them to determine their positions in the periodic table.

a) Given that sulphur has the configuration, $1s^2 2s^2 2p^6 3s^2 3p^4$:

i) Identify its period. (1 mark)

ii) Identify its group. (1 mark)

b) Potassium has the configuration, $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$:

i) Identify its period. (1 mark)

ii) Identify its group. (1 mark)

c) Name the chemical family of potassium. (1 mark)

23. Learners studied ion formation.

a) State the octet rule. (2 marks)

b) Distinguish between cation and anion. (2 marks)

24. Electron arrangement influences chemical behaviour.

a) Explain its role in the periodic table. (1 mark)

b) Explain why elements in the same group have similar properties. (1 mark)

c) Explain how it helps in writing chemical formulae. (2 marks)

25. Learners discussed the importance of chemistry in daily life.

a) Outline four applications of chemistry. (4 marks)

- i. _____
- ii. _____

b) Name two industries where chemistry is applied. (2 marks)

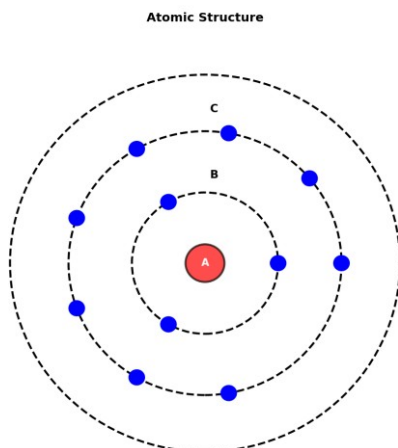
- i. _____

SERIES 9

SECTION A (40 MARKS)

Answer ALL questions in this section.

1. The diagram below shows part of an atom. (5 marks)

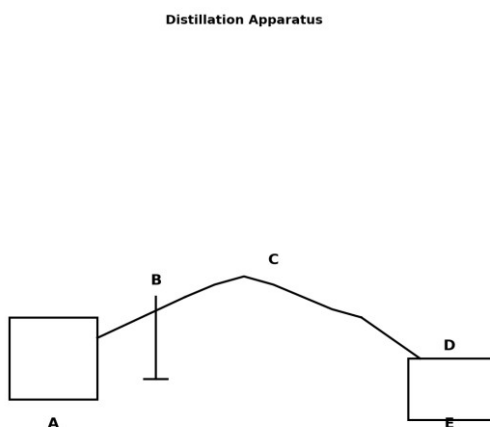


(a) Identify the parts labeled A, B and C in the diagram above. (3 marks)

(b) State the relative mass of a proton. (1 mark)

(c) Define isotopes. (1 mark)

2. The diagram shows a laboratory apparatus used in chemistry. (6 marks)

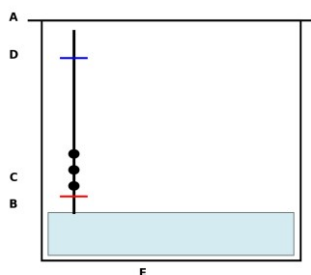


(a) Name the apparatus labeled A, B, C, D and E. (5 marks)

(b) What is the purpose of apparatus C? (1 mark)

3. Study the diagram below and answer the questions that follow. (7 marks)

Paper Chromatography



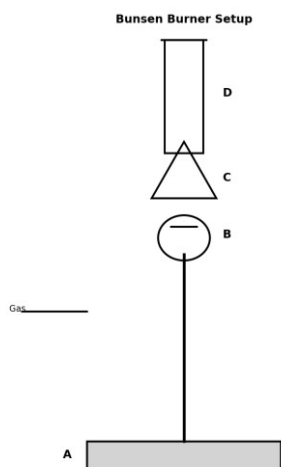
(a) Name the apparatus labeled A and D. (2 marks)

(b) What do B and C represent? (2 marks)

(c) State one use of paper chromatography. (1 mark)

(d) Calculate the R_f value of the spot if distance from origin to spot is 3 cm and distance to solvent front is 5 cm. (2 marks)

4. The diagram below shows an experimental setup for heating. (8 marks)

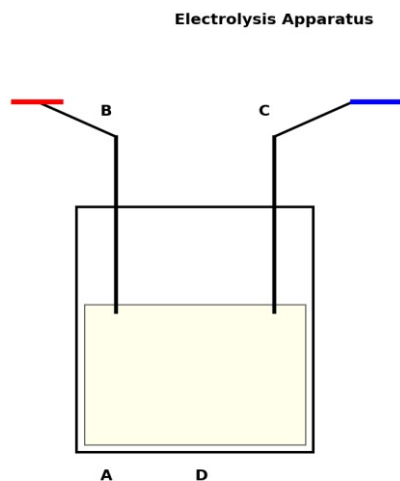


(a) Identify the apparatus labeled A, B, C and D. (4 marks)

(b) Explain the function of apparatus C. (2 marks)

(c) State two safety precautions when using this apparatus. (2 marks)

5. The diagram below shows an electrolysis cell. (8 marks)



(a) Identify the parts labeled A, B and C. (3 marks)

(b) Explain what happens at the cathode during electrolysis. (2 marks)

(c) Write an equation for the electrolysis of molten sodium chloride. (3 marks)

SECTION B (40 MARKS)

Answer ALL questions in this section.

6. Calculate the molar mass of: (a) Calcium carbonate, (b) Sodium hydroxide, (c) Sulphuric acid. (6 marks)

7. A sample of magnesium oxide has molar mass 40 g/mol. Calculate the number of moles in 8 g. (3 marks)

8. What volume of hydrogen gas would be produced when 1.2 g magnesium reacts with dilute hydrochloric acid at RTP? (5 marks)

9. Calculate the percentage composition by mass of calcium in calcium carbonate. (6 marks)

[Atomic masses: Ca = 40, C = 12, O = 16]

10. Differentiate between evaporation and distillation. (4 marks)

11. Describe the method of preparing a pure solid by crystallization. (6 marks)

12.(a) Define oxidation in terms of oxygen. (2 marks)

(b) State two sources of oxygen in the laboratory. (2 marks)

(c) Describe how you would test for oxygen gas in the laboratory. (2 marks)

13.(a) Define a chemical reaction. (2 marks)

(b) Give three types of chemical reactions with examples. (6 marks)

SERIES 10

1. Define the following terms(3marks):

i)cation

ii)anion

iii) drug

2. State one application of chemistry in each of the following fields (3marks)

i)Agriculture

ii)Clothing industry

iii)Medicine

3. Describe the contributions of the following scientists to the development of the atomic theory (3 marks)

i)John Dalton

ii) J.J Thompson

iii) Ernest Rutherford

4. Lithium has 2 isotopes with mass numbers 6 and 7 respectively. If the R.A.M of lithium is 6.94, calculate the percentage abundance of each isotope (3 marks)

5. An element P has two isotopes with relative abundance 65% and 35%. If the mass number of the two isotopes is X and 31 respectively, find the mass number represented by X given that the R.A.M of P is 30. (3 marks)

6. Write the electron arrangement of the alkali metals using the s and p notation (3 marks)

7. Explain how electron spin contributes to the stability of the atom (2marks)

8. Briefly explain the contribution of the following scientists to the development of the periodic table (2 marks)

i) Dmitri Mendeleev

ii) Henry Moseley

9. Using dots(.) and cross(X) to represent electrons show how the following atoms form ions (6 marks).

i) Aluminium

ii) Oxygen

iii) Fluorine

10. Derive the chemical formulae for the following compounds (6 marks)

i) Aluminium chloride

ii) Ammonium sulphate

iii) Magnesium nitride

iv) Silicon (iv) oxide

v) Sulphur (iv) oxide

vi) Nitrogen (ii) oxide

11. Write and balance the chemical equations for the following reactions (4 marks)

a) Iron + Sulphur $\rightarrow$ Iron (ii) sulphide

b) Sodium hydroxide + nitric (v) acid $\rightarrow$ Sodium nitrate + water

12. Balance the following equations: (12 marks)

(a) $\text{Mg(s)} + \text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$

(b) $\text{Na(s)} + \text{H}_2\text{O(l)} \rightarrow \text{NaOH(aq)} + \text{H}_2\text{(g)}$

(c) $\text{NaOH(aq)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{Na}_2\text{SO}_4\text{(aq)} + \text{H}_2\text{O(l)}$

(d) $\text{CuCO}_3\text{(aq)} + \text{HNO}_3\text{(aq)} \rightarrow \text{Cu(NO}_3)_2\text{(aq)} + \text{CO}_2\text{(g)} + \text{H}_2\text{O(l)}$

(e) $\text{H}_2\text{S(g)} + 3\text{O}_2\text{(g)} \rightarrow \text{SO}_2\text{(g)} + \text{H}_2\text{O(l)}$

(f) $\text{C}_2\text{H}_6\text{(g)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)} + \text{H}_2\text{O(l)}$